

PRACTICAL WORKSHEET 2: Comparing groups

This week we will test some scientific hypotheses using statistics. Specifically, we will focus on a type of test that is used to compare means: the t-test. The test examines whether our estimate of a mean for a group differs from a prior expectation, or from the mean of another group. It's a fundamental statistical test that can help you assess whether two groups are different from each other statistically.

There are three types of t-test we shall look at

- 1) A **single sample t-test**: which compares your sample mean against an expected mean (i.e. tests whether your sample is likely to come from the population that is expected)
- 2) The **independent samples t-test**: which compares the means of two sample groups (i.e. tests whether the two groups come from the same or different populations)
- 3) The **paired t-test**: which compares the mean values of the same group of individuals under two different treatments (i.e. tests whether the population under treatment A is different from the same population under treatment B)

We will also investigate a fourth test, used when t-tests aren't appropriate, which is a non-parametric test of medians – the **Wilcoxon-Mann-Whitney test**

As with the previous worksheet, the blue boxes show you a sample of the datasets that you need to load into R and the black boxes will instruct you on the actions you need to within R to perform the required calculations.

By the end of this worksheet you should be able to:

- 1) Get R commander to carry out three different types of t-test and the non-parametric test
- 2) Interpret the results of the above tests in a meaningful manner
- 3) Understand how to make diagnostic tests to confirm that your data meet the statistical assumptions of t-tests

The assessed question you will submit will test how well you have achieved these goals.

Question 1: single-sample t-test

It has been suggested that flower temperature can influence the likelihood of being pollinated by bees. Norgate et al (2010) sought to test this by investigating the visitation rates to artificial feeders (filled with sucrose solution) of Australian native stingless bees, *Trigona carbonaria*. They presented individual bees with two feeders, one at ambient temperature (23°C), and one six degrees warmer (29°C). They observed how long the bees spent at each feeder, and split this into a percentage time divided between the two feeders (i.e. if bees spent 40% time at the ambient temperature feeder they spent 60% of time at the warmer feeder).

Format of sle251beeprefs data file

BeelD	pctambient
AA	45
AB	39.3
AC	32.3
AE	34.3
AF	35.6
	..

BeelD The identification code of the individual bee

pctambient The percentage of time spent at the ambient temperature feeder

Open the *sle251beeprefs* file in *R commander*.

To open a dataset within R

Data**Import data****From Excel file**

Under **Enter name for data set:** type a name for your data

select **OK**

Locate the file on your computer and select it

Press the **View data set** button to see your data

NOTE for those who have downloaded R commander onto their personal computer: the most recent version of R commander on PCs may have a problem reading xlsx files. You may have to upload your data file in the CSV (comma separated value) format. You can do this by selecting the csv file and following these instructions

INSTRUCTIONS for uploading a CSV file

Data**Import data****from text file, clipboard, or URL...**

Under **Enter name for data set:** type a name for your data

Under the **Field separator** menu, select **Commas**

select **OK**

Locate the file on your computer and select it

Press the **View data set** button to see your data

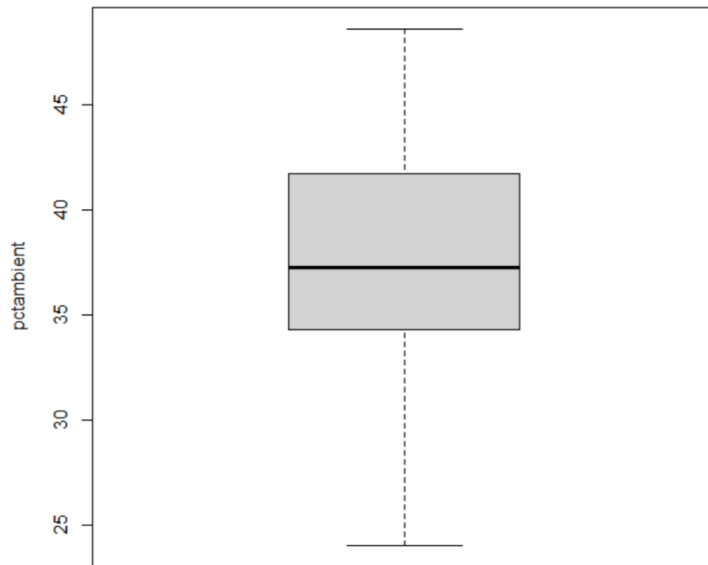
a) What is the population being sampled?

- the population being sampled was the Australian native stingless bees.

b) How many bees were sampled (what is the sample size)?

- 30 bees

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Construct a box plot for *pctambient* (see instructions below)



c) What is the shape of the distribution? Is it symmetrical or skewed? Does this suggest the distribution is normal or non-normal?

- The distribution is more-or-less symmetrical (with very little skew), suggesting normality.

To make a boxplot within R

Graphs

Boxplot

Select *pctambient* from Variable list

OK

d) What is the mean and standard deviation for the percentage of time bees spent at the ambient temperature feeder (enter in the table below)?

To calculate summary statistics in R

To get means and standard deviation:

Statistics

Summaries

Numerical summaries...

Choose *pctambient* from Variables list.

Under the **Statistics** tab you can deselect **Interquartile range** and **quantiles**

OK

Summary statistic	pctambient
Mean	37.73
Standard deviation	5.319784

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Now test whether the observed percentage of time spent at the ambient temperature feeder differs from the expectation that bees would spend an equal amount of time at each feeder. If there is no active preference for nectar at a particular temperature, and bees visit feeders just by chance, then the mean percentage of time spent at the ambient feeder should be 50% (i.e. the mean percentage expectation is 50)

Performing a single-sample t-test in R

Statistics

Means

Single sample t-test

Pick *pctambient* as the Variable

Leave Population mean != mu0 checked

In the box next to *Null hypothesis: mu =* type **50**

OK

e) What is the null hypothesis being tested here?

- The null hypothesis being tested is that if there is no active preference for nectar at a particular temperature, then the mean percentage of the ambient percentage would be 50%.

f) Summarize the results as if you were writing a research report (delete the wrong answers):

The mean percentage of time spent by bees at the ambient temperature feeder was **37.73**

significantly lower than

that expected by chance ($t = -12.633$, $df = 29$, $P = 2.572e-13$).

g) What is the correct interpretation of these results? Do they support the hypothesis that bee visitation to flowers is influenced by temperature?

- These results do not support the hypothesis, as the mean percentage of the ambient percentage was 37.73%. This suggests that bee visitation may be influenced by temperature.
- Bees spend significantly less time at the ambient temperature feeder than would be expected by chance, indicating they prefer nectar at a warmer temperature than ambient. So, yes, this supports the hypothesis that visitation to flowers is influenced by temperature.

Question 2: independent samples t-test

It is thought that freshwater invertebrates are more prone to parasitic infections when they are suffering from nutrient deficiency (as can happen in polluted waterways). To test this Pulkkinen et al. (2014) investigated numbers of a sporidian parasite, *Glugoides intestinalis*, in the digestive tracts of *Daphnia magna* water fleas. The *Daphnia* were fed diets of one of two treatments, the first where the food was phosphorus limited, and the second where the phosphorus was at sufficient levels. After 32 days of these treatments they dissected the *Daphnia*, and counted the number of parasite spores in the digestive tract.

Open the *sle251parasites* file in *R commander*.

Format of sle251parasites file

phosphate	parasites
Low	170
Low	610
..	..
Normal	250
Normal	64
..	..

phosphate The diet treatment to which the *Daphnia* was subjected

parasites The number of *Glugoides* parasite spores in the digestive tract

Note the structure of the data file with one column identifying the group and another containing the values of the response or dependent variable, and each row being one individual patient (observation). This is a standard structure for data files for statistical analyses of this type – the variables are in different columns and each replicate observation is on a different row.

a) What is the null hypothesis (H_0) being tested in this study?

- There is no difference in the mean number of parasite spores in the digestive tracts of *Daphnia magna* water fleas between those fed a phosphorus limited diet and those fed a phosphorus sufficient diet.

b) What is the dependent or response variable?

- The number of parasite spores present in the digestive tracts of the water fleas.

c) What is the grouping (or independent or predictor) variable?

- The grouping variable in this study is the level of phosphorus in the diet /

d) What is/are the population(s) being sampled?

- The population being sampled in this study is *Daphnia magna* water fleas on the low and normal phosphorus diets.

The appropriate statistical test for testing the hypothesis about differences between the means from two independent populations is an independent samples *t*-test. Like any statistical test, there are some key assumptions of a *t* test. These are listed – but indicate how you might check if these assumptions are met?

Assumption	Diagnostic check
1. Normal distribution to the data	1) use box plots - look for indications of asymmetry in distributions 2) use frequency histograms
2. Equal variances in the two groups (Homogeneity of variances)	1) use box plots 2) compare variance ratio for the 2 groups - if the ratio is greater than 2, variances are not homogeneous 3) Levene's test
3. Independence of observations	- Sampling is random and data points are independent of each other. - Think about the set up of the experiment – is there pseudo-replication?

e) Determine the mean and variance for parasite counts for the *Daphnia* in the two diet treatments.

Summary statistic	Low	Normal
Mean	565.0385	241.3333
Variance	126450.44	24876.33

To calculate mean and variance in R

Statistics

Summaries

Table of statistics...

Choose *parasites* from *Response variables* list.

Choose *phosphate* from *Factors* list

Choose *mean* as the statistic to report

For variance: choose *Other(specify)* and type

var

in the adjacent box

OK

Check the two main assumptions (normality and equal variances):

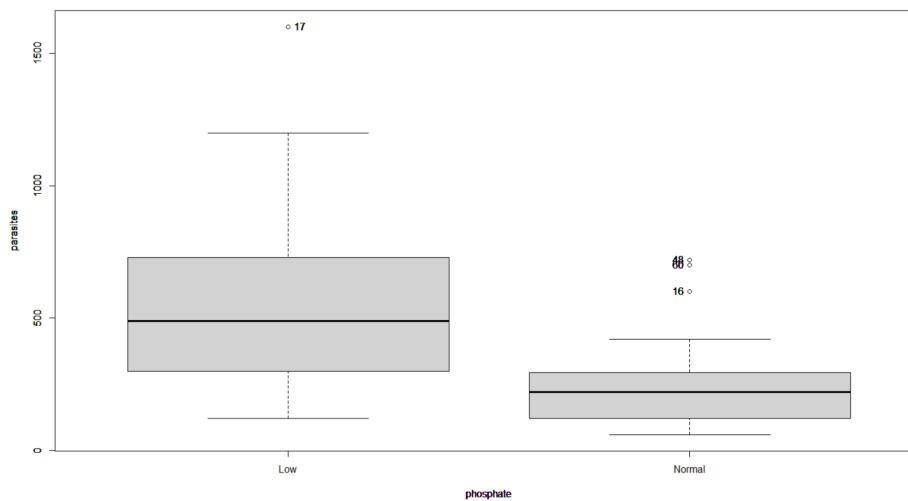
f) Compare the variances in the parasite counts for each group – are they similar? Is the ratio of variance (larger variance:small variance) greater than 2:1? If so this an indicator that the variances are not equal

$$\text{Ratio} = 126450.44 / 24876.33 \approx 5.08$$

Since the ratio of the “low” group variance to the “normal” group variance (5.08) is greater than 2:1, this suggests that the variances are not equal. Thus, the variance of the "Low" group is substantially larger than the variance of the "Normal" group.

g) Construct a boxplot of the parasite counts for each group – is there any evidence of skewness to the data?

Box Plot w/ Outliers



The data appear to be slightly, but not heavily, skewed. Although the main distributions are roughly symmetrical, both groups have large outliers at the top of the data range, skewing the data.

To create a boxplot by groups in R

Graphs

Boxplot

Select *parasites* from *Variable* list

Click *Plot by groups...*

Select *phosphate* as grouping variable (factor)

OK

OK

h) based on the above checks, have the assumptions of an independent samples t-test been met?

NO, there is observable asymmetry in the distributions and the variance ratio indicates that the variances are not homogenous. Thus, the assumptions of an independent samples t-test have not been met.

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Try log-transforming the parasites data.

Log transforming data in R

Data

Manage variables in active data set

Compute new variable

Type in name of new transformed variable in *New variable name* box (e.g. **logparasites**)

Type in **log10(parasites)** in *Expression to compute* box.

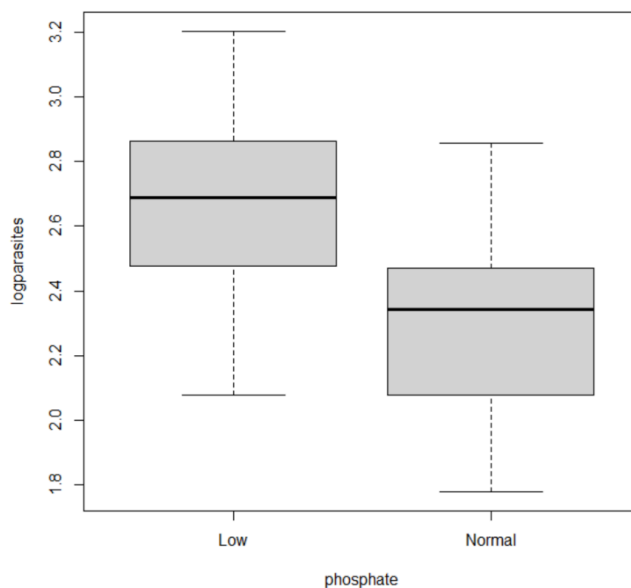
OK

Click **View data set** to check new column has appeared correctly

i) For the log-transformed **logparasites** variable determine the mean and variance using the procedures described earlier.

Summary statistic	Low	Normal
Mean	2.665	2.299
Variance	0.086	0.077

Also construct a boxplot of the **logparasites** data by group



j) Does the boxplot indicate that the **logparasites** data better fit a normal (symmetrical) distribution?

Yes, the logparasites data better fits a normal symmetrical distribution and the outliers have also disappeared.

k) Based on the results above, would the **parasites** or the **logparasites** data better comply with the assumptions of an independent-samples t-test?

The logparasites data would be better. They conform better to normality, and they also have more equal variance (variance ratio 1.12: 1)

Now carry out an independent samples t-test to test the null hypothesis.

t-tests

There are two versions of an independent samples t-test:

- The “equal variances t-test” is the usual t-test – this assumes that the variances are equal and is the better and more robust test to use.
- The “unequal variances t-test” adjusts the *df* (degrees of freedom) downwards to make the test more conservative and doesn’t require equal variances in the two groups, but is usually only used where the variances can’t be easily adjusted to meet the requirement of equal variance.

If the assumptions of normality and homogeneity of variances seem reasonable, use the former. If the variances are very different (e.g. ratio of two variances $\geq 2:1$), use the latter.

To run an independent samples t-test in R

Statistics

Means

Independent samples t-test...

Add *parasites* **or** *logparasites* (decide from above which is more appropriate) into *Response Variable* box
Choose *phosphate* from *Groups* list
Leave *Alternative Hypothesis* as *Two-sided*
Choose whether to assume equal variances or not (see note above about two different sorts of t-test).
OK

If there was no issue with normality the unequal variance t-test could be used here, however, the log-transformed data conform better with normality, so better to use them and the equal variance t-test.

a) Write out the results as though you were writing a research report (delete the wrong answer):

The mean number of parasites in digestive tract of *Daphnia* given a low phosphate diet

Significantly greater than

the mean number of parasites in *Daphnia* fed a normal diet ($t = 5.091$, $df = 63$, $P = 3.466e-06$).

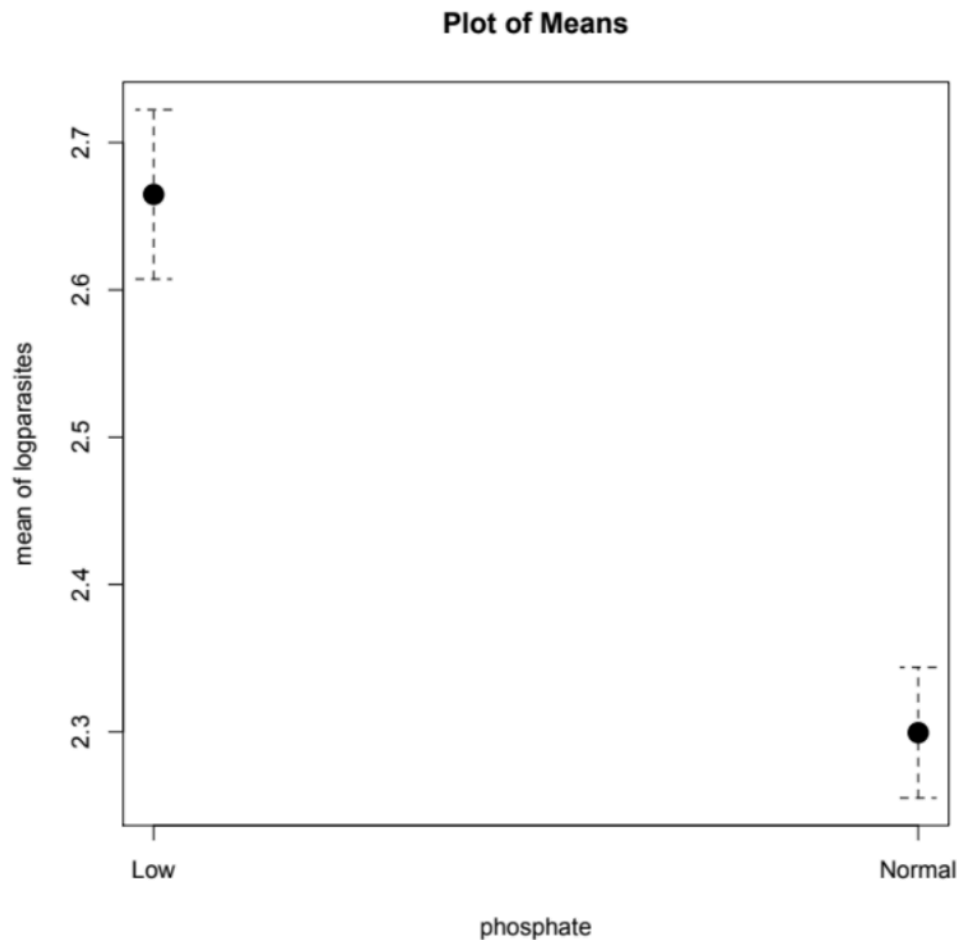
b) What would be your conclusion in regard to whether *Daphnia* on nutrient deficient diets are more susceptible to parasitic infection?

Yes, there is evidence that nutrient-deficient *Daphnia* are more likely to be heavily infested with parasites

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c) Such a result would normally be associated with a graph to show the means and some measure of variation or precision. Construct plot of means in R showing the parasite counts (± 1 standard error). Remember to use the same version of the data (**parasites** or **logparasites**) as you used for the t-test. Note: to insert a plot into the worksheet save as a pdf or jpeg and paste into the document below.

Insert your plots of means here:



To create a plot of means in R

Graphs

Plot of means...

Select *phosphate* from *Factors* list

Select *parasites* or *logparasites* (as appropriate) from *Response variable* list

Under the Options tag for *Error Bars* select *Standard errors*

Also under the Options tag de-select *Connect profiles of means* so it is not ticked (makes output look nicer!)

OK

Question 3: Paired *t*-test

Is a stay in the intensive care unit (ICU) of a hospital more stressful for patients or for their close family relatives? To investigate this Fumis et al. (2015) interviewed patients who had recently been admitted to the ICU (assuming they were in a condition where they could respond), as well as one of their close relatives (parent, sibling). They assessed their answers on the HADS scale (Hospital Anxiety and Depression Scale), focusing particularly on their anxiety scores. Their results are in the data file *sle251hospital*

Format of *sle251hospital* data file

Family	patient	relative
A	5.6	8.3
B	6.6	7.3
C	5.8	7.0
..	..	..

Family The coded identity of the family in the study

patient The HADS anxiety score for the patient

relative The HADS anxiety score for the relative

Open the *sle251hospital* file in *R* following the procedures detailed in question 1.

a) Explain why a paired *t*-test is the appropriate statistical test for this study

Because the two groups are not independent of each other, being members of the same family and sharing circumstances in regard to the reason for being in hospital.

Note: for paired *t*-tests, the two categories appear as separate columns rather than as a single categorical variable. Compare the structure of this dataset (paired *t*-test) set with that of the previous (standard *t*-test) data set.

b) What is the null hypothesis in this study? Under this null hypothesis what is the expected mean difference between the two treatments?

There is no difference in the anxiety levels of patients and their relatives when they are in the ICU.

The expected mean difference between the two groups is zero.

To run a paired t-test in R

Statistics

Means

Paired t-test...

Select first and second variables in pair (e.g. *patient* and *relative*).

Leave *Alternative Hypothesis* as *Two-sided*

OK

c) Use a paired *t*-test to test whether there is a difference in anxiety scores for patients and their relatives. Write the results out as though you were writing a research report (see questions 1 and 2 for examples). Remember to include the *t*-value, degrees of freedom and *p* value in your description.

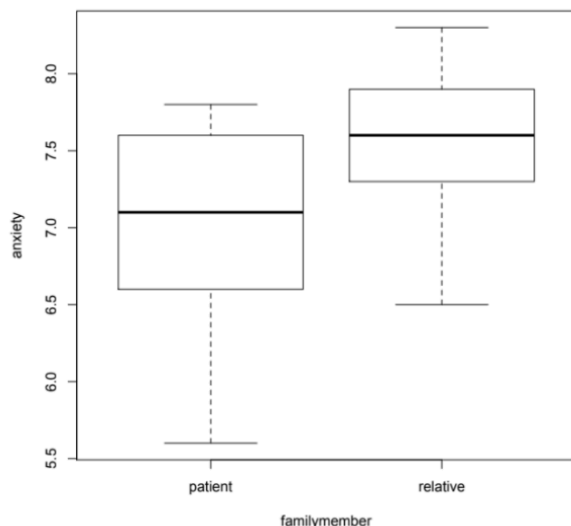
The mean anxiety levels of patients was not significantly different from the mean anxiety levels of their relatives ($t = 1.858$, $df = 12$, $P = 0.0879$)

d) what would be the appropriate conclusion from these results?

That stays in ICU are equally as stressful for patients as for their relatives.

NOTE: that the *P* value is very close to significance, and there is a trend that relatives are more stressed than the patients – however at this point in time it is better to conclude that there is no difference

e) Create a box plot to compare the anxiety scores of patients and their relatives and paste the plot below



To create a box plot for paired data in R

Before plotting we need to reshape the data to the format required for creating boxplots

Data

Active data set

Stack variables in active data set

Select the **patient** and **relative** variables

Name your stacked dataset (e.g. *stress*)

Name your variable (e.g. *anxiety*)

Name your factor (e.g. *familymember*)

Now create the boxplot using the new data set

Graphs

Boxplot

Select *anxiety* from *Variable* list

Click *Plot by groups...*

Select your factor (*familymember*) as grouping variable

OK

OK

Question 4: Non-parametric tests

Sometimes the assumption of a normal distribution for our data cannot be met - the data are too oddly distributed or the sample sizes are too small to evaluate the normality assumption adequately. Under these circumstances, a non-parametric test might be appropriate where the ranks of the data are analysed rather than the original values.

Wallach et al (2009) were investigating whether control measures such as poison baiting and shooting had been successful in reducing dingo numbers at various locations in the South Australian arid zone. They assessed dingo numbers indirectly by counting the number of scats (droppings) as they walked a 500m transect. They did this on seven stations, three where there had been shooting/baiting, and four where there had been no such control measures

The data file is *sle251dingoscat* – load it

Format of sle251dingoscat data file

Site	Control	Scats
Red Lake	No	6
Andamooka	No	13
Nantawarrinna	No	11
..	..	..

Site The name of the station where the transect was conducted

Control Whether there had been control measures against dingos at that station (yes or no)

Scats The number of dingo scats observed along a 500m transect at that station

a) The Wilcoxon-Mann-Whitney test is a non-parametric test for comparing two groups. What null hypothesis does this test evaluate (hint: how does it differ from a t-test)?

It tests the null hypothesis that the two population medians (nb. medians not means) are equal

b) Why is a non-parametric test appropriate in this instance rather than a t-test?

The sample sizes for each group are very small (4 and 3). There's no reliable means of testing the assumption of normality, so the Wilcoxon-Mann-Whitney test is a safer bet.

Test the null hypothesis of no difference in the number of dingo scats found at the two different types of site using a Wilcoxon-Mann-Whitney test.

To run a non-parametric Wilcoxon-Mann-Whitney test in R

Statistics

Nonparametric tests

Two-sample Wilcoxon test

Under **Groups** select **Control**

Under **Response Variable** select **Scats**

OK

Note: to compare the trends you need to compare the medians for the two treatments. The medians are given under the first part of the output above the W-statistic output.

c) Identify the following from the output;

Median number of scats (no controls)	10
Median number of scats (controls in place)	1
Mann-Whitney W statistic	12
<i>P</i> value	0.04975

d) Based on this output, what would be the most appropriate conclusion?

There is a significant difference in dingo abundance between sites where control measures have been put in place compared to sites where no control is present. Specifically, there is greater abundance of dingos at sites without control measures.